A Central Processing Unit Architecture for Native Parametric and Symbolic Computation

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Title: A Central Processing Unit Architecture for Native Parametric and Symbolic Computation

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Cross-Reference to Related Applications

This application is related to the inventions described in the U.S. Patent Applications for "Method and System for High-Radix Data Representation Using Parametric Mathematical Encoding," "High-Radix-Number-Representation," and "High-Precision-Fractional-Number-Representation". The contents of these applications are hereby incorporated by reference in their entirety, as the present invention provides a processing architecture uniquely suited to operate on the data representations described therein.

Abstract

A Central Processing Unit (CPU) architecture configured for the native storage and manipulation of numbers represented as parametric functions or symbolic 'recipes'. The architecture comprises a set of parametric registers capable of holding a data structure that defines a number's mathematical formula, rather than its evaluated, fixed-width binary value. The

CPU further includes a Parametric Logic Unit (PLU), which executes instructions on these registers by performing symbolic mathematical operations—such as combining two parametric representations—rather than traditional binary arithmetic. This design allows the CPU to perform calculations at a precision limited only by its internal logic, not by the fixed bit-width of its registers or data bus, eliminating the loss of precision that occurs when high-precision numbers are evaluated prematurely for processing.

Background of the Invention

Conventional CPU architectures are founded on the principle of operating on fixed-width binary representations of numbers. Integers and floating-point values are stored in registers of a specific size (e.g., 64 bits), and Arithmetic Logic Units (ALUs) perform binary operations on these discrete values. While this model has been incredibly successful, it has a fundamental limitation: any number that cannot be perfectly represented within the fixed bit-width (such as irrational numbers or high-precision fractions) must be truncated or approximated. When data is transmitted with greater precision, as in a parametric encoding system, that precision is immediately lost upon being loaded into a conventional CPU register for processing. All subsequent computations are performed on this less-precise, approximated value.

Summary of the Invention

The present invention is a CPU architecture that natively processes numbers in a parametric, symbolic form. Instead of approximating a number to fit into a register, the CPU stores the number's exact mathematical "recipe." This fundamentally changes the nature of computation from numerical evaluation to symbolic manipulation at the hardware level.

The architecture includes two key innovations. First, it replaces traditional general-purpose registers with 'parametric registers'. Each parametric register is a hardware data structure designed to store the components of a mathematical expression, such as operands, operators, and coefficients of a series. For example, the constant Pi is not stored as '3.14159...'; it is stored as a symbol 'Pi' which the CPU hardware natively understands.

Second, the traditional ALU is supplemented or replaced by a Parametric Logic Unit (PLU). The PLU operates on the contents of the parametric registers. When instructed to add two numbers, the PLU does not add their binary values; it creates a new parametric representation that is the symbolic sum of the two input "recipes." A new instruction set architecture

(ISA) provides instructions for loading, storing, and manipulating these parametric representations (e.g., ‘P_LOAD‘, ‘P_ADD‘, ‘P_MULTIPLY‘). An evaluation instruction (‘P_EVAL‘) is also provided to compute the final numerical result of a parametric representation to a specified precision when an actual value is needed for output.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram contrasting a traditional CPU architecture with the parametric CPU architecture of the present invention.

FIG. 2 is a conceptual diagram illustrating the structure of a parametric register holding the mathematical "recipe" for a number.

FIG. 3 is a flowchart illustrating the execution of a parametric addition (‘P_ADD‘) instruction within the Parametric Logic Unit.

Detailed Description of the Invention

Referring to **FIG. 1**, a traditional CPU (100) is compared with the parametric CPU (150). The traditional CPU (100) contains standard registers (110) that store fixed-width binary values, which are operated on by an ALU (120). The parametric CPU (150) introduces parametric registers (160), capable of storing structured symbolic data. These registers are operated on by a Parametric Logic Unit (PLU) (170), which performs symbolic manipulation according to a new parametric instruction set.

FIG. 2 illustrates a parametric register (200) in more detail. Unlike a simple array of bits, the parametric register is shown holding a data structure (210) representing the fraction '7/16'. The structure has distinct fields for an operand '7' (220), an operator 'divide' (230), and another operand '16' (240). This allows the CPU to internally store the true concept of the number, preserving its full precision indefinitely. A notable constant like Pi could be stored as a single, irreducible symbol 'Pi' in a similar structure.

FIG. 3 shows a flowchart (300) for a parametric 'add' operation. The process begins when the CPU decodes a 'P_ADD' instruction targeting two source registers, R1 and R2 (310). The PLU fetches the parametric "recipes" from R1 and R2 (320). Instead of computing a numerical result, the PLU creates a new data structure representing the symbolic sum of the two inputs (e.g., a new tree with '+' at the root and the R1 and R2 recipes as children) (330). This new parametric representation is then written to the destination register, R3 (340). The entire operation is performed without any loss of precision, as no numerical evaluation occurs.

Claims

What is claimed is:

1. A central processing unit, comprising:
    - a plurality of parametric registers, wherein each parametric register is configured to store a data structure representing a mathematical expression of a number; and
    - a Parametric Logic Unit (PLU) communicatively coupled to the parametric registers, the PLU configured to execute instructions by performing symbolic mathematical operations on the data structures stored in said registers.
2. The CPU of claim 1, wherein the data structure represents a number as a sequence of operands and operators.
3. The CPU of claim 1, wherein the data structure represents a mathematical constant as a native symbol.
4. The CPU of claim 3, wherein the mathematical constant is Pi or Euler's number.
5. The CPU of claim 1, wherein the PLU, in executing an arithmetic instruction on a first and a second parametric register, is configured to generate a new data structure

representing the symbolic combination of the expressions from the first and second registers.

6. The CPU of claim 1, further comprising an instruction decoder configured to interpret a specialized instruction set architecture including instructions for loading, storing, and symbolically manipulating the data structures in the parametric registers.

7. The CPU of claim 6, wherein the instruction set includes an evaluation instruction that causes the PLU to compute a numerical value from a data structure to a specified precision.

8. A method for computation within a central processing unit, comprising:

   - storing a first number as a first data structure representing a first mathematical expression in a first parametric register;
   - storing a second number as a second data structure representing a second mathematical expression in a second parametric register; and
   - executing an instruction by creating a third data structure representing the symbolic mathematical combination of the first and second expressions, and storing said third data structure in a third parametric register.

9. The method of claim 8, wherein the execution of the instruction occurs without evaluating the first and second mathematical expressions to fixed-precision numerical values.

Drawings

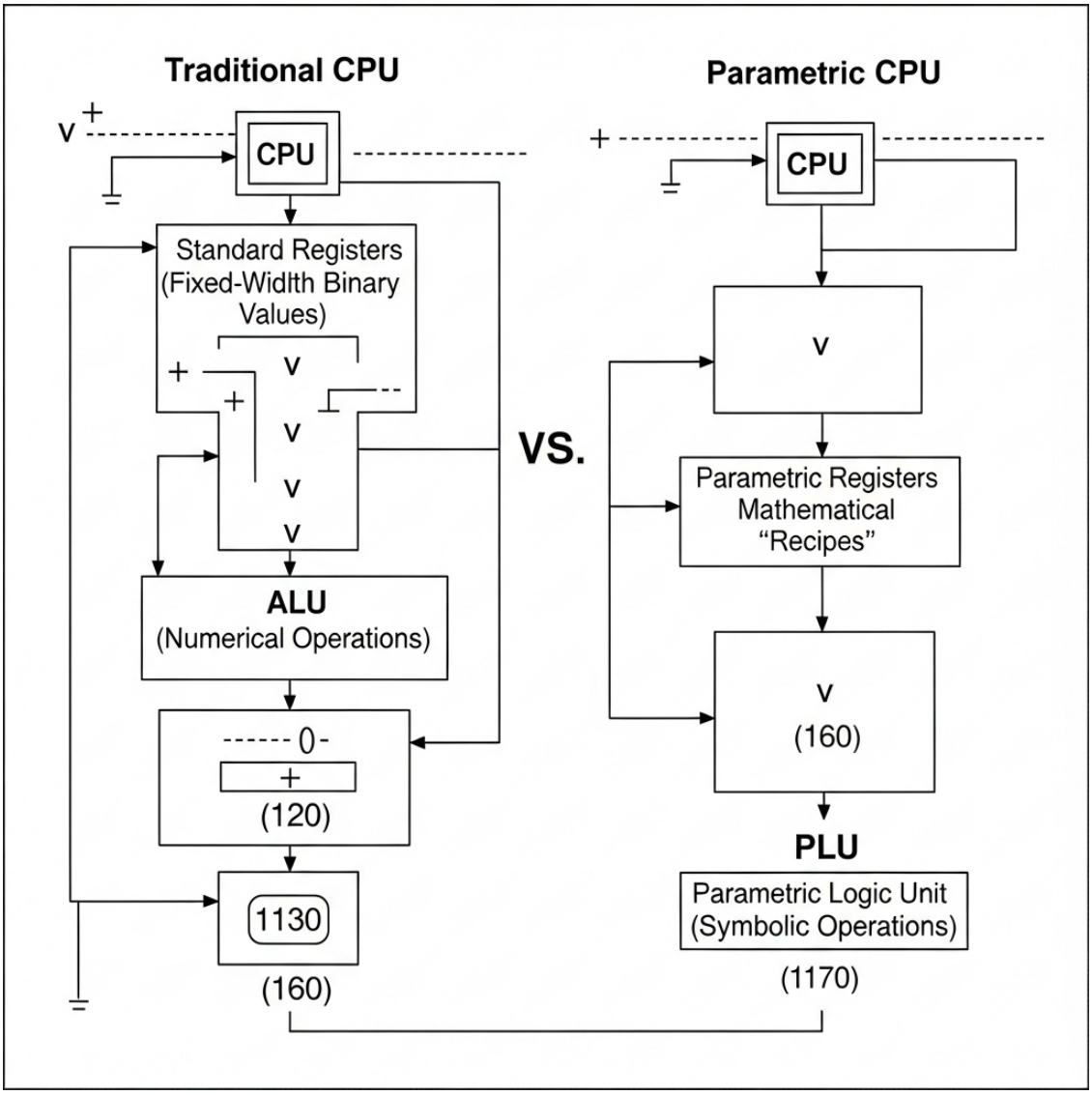


Figure 1: Block diagram contrasting a traditional CPU architecture with the parametric CPU architecture of the present invention.

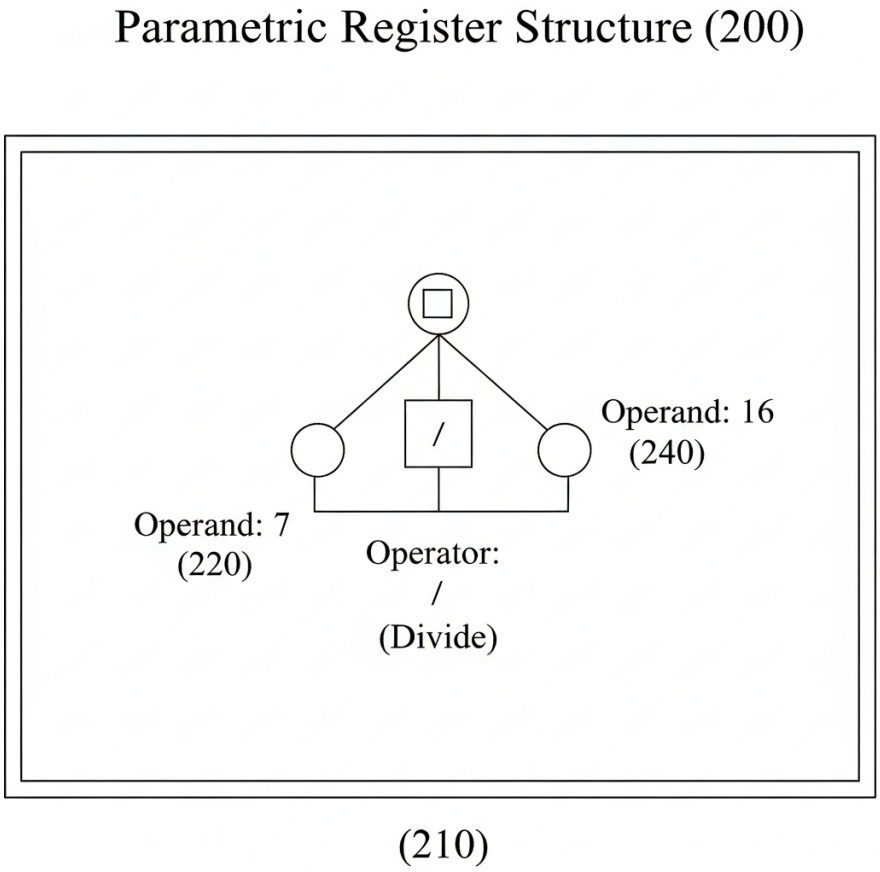


Figure 2: Conceptual diagram illustrating the structure of a parametric register holding the mathematical "recipe" for a number.

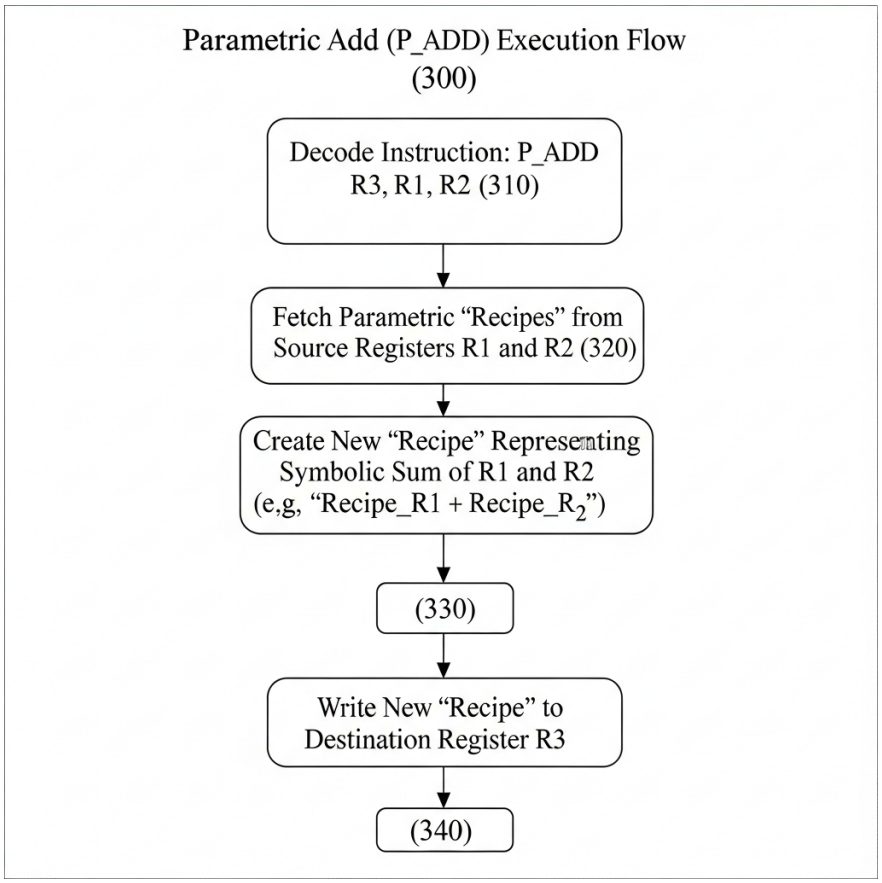


Figure 3: Flowchart illustrating the execution of a parametric addition (P_ADD) instruction within the Parametric Logic Unit.